Birth of “Foundation Models”: Learning from Prompts and Reasoning

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University of Colorado Boulder
Spring 2026

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https://dannagurari.colorado.edu/course/neural-networks-and-deep-learning-spring-2026/

Review

- Last lecture:
  - The quest for growth: the secret sauce for success
  - Scaling laws: predictable gains from model size, data, & compute
  - Fueling scale: harnessing the Internet as training data
  - Revelation: few-shot and zero-shot abilities emerge
  - Programming tutorial

- Assignments (Canvas):
  - Lab assignment 3 due in 1 week
  - Final projects – finding partners (speed virtual “dating” event on Monday)

- Questions?

Today’s Topics

- From specialists to generalists: birth of “foundation” models
- Generalizing to any task: the art of prompting
- Thinking slower: reasoning for better outcomes
- Beyond text: prompting in other modalities
- Prompting activity

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Scaling Up Changed the Landscape: After Zero-Shot Success Comes Generalist Models

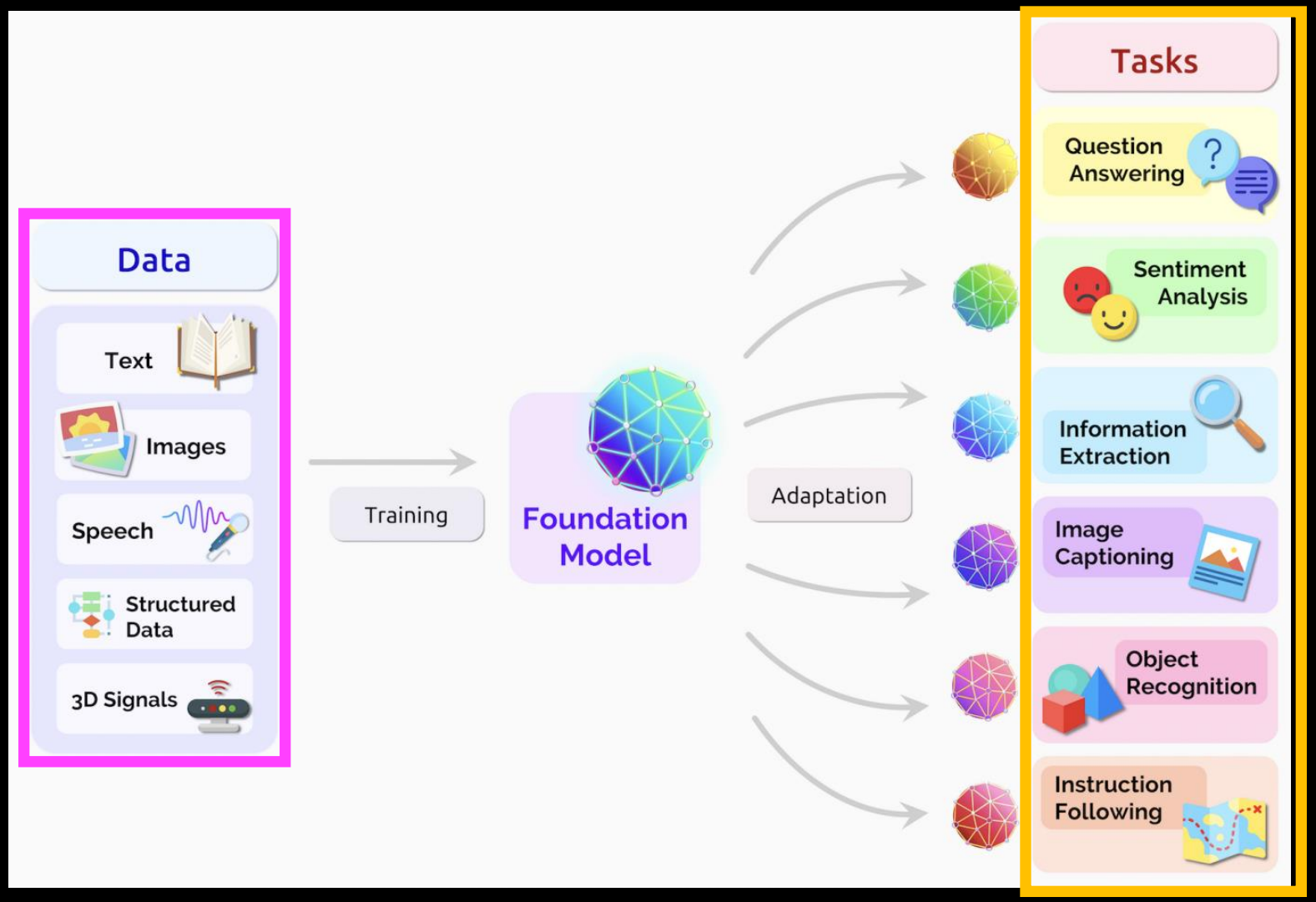


VS



https://www.esquire.com/lifestyle/a41044437/7-unconventional-uses-for-your-swiss-army-knife/

Approach: Large-Scale Training and Evaluation



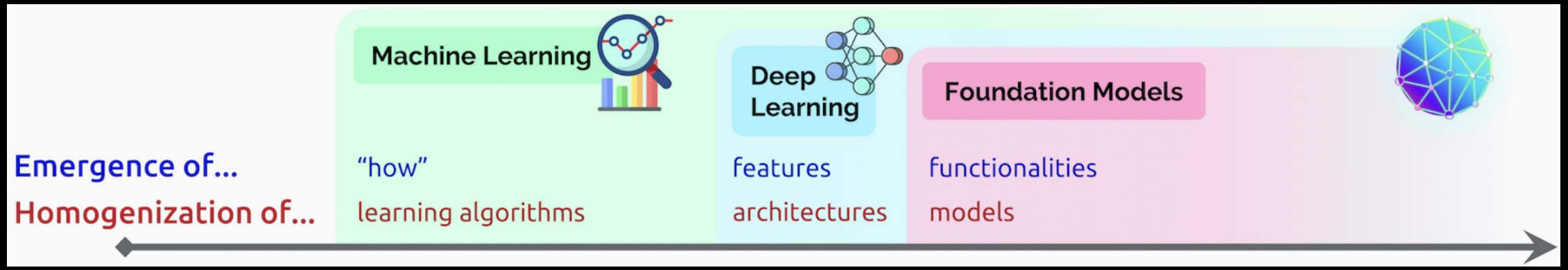
Evaluated on benchmark datasets for many:

1. Tasks (e.g., object recognition, scene classification)

2. Data distributions for the same task (e.g., ImageNet vs data from blind people)

Bommasani et al. On the Opportunities and Risks of Foundation Models. arXiv 2021.

New Paradigm: “Foundation” Model



The term was coined in 2021 to account for the paradigm shift where a single model could now support many downstream tasks as is by just using prompting!

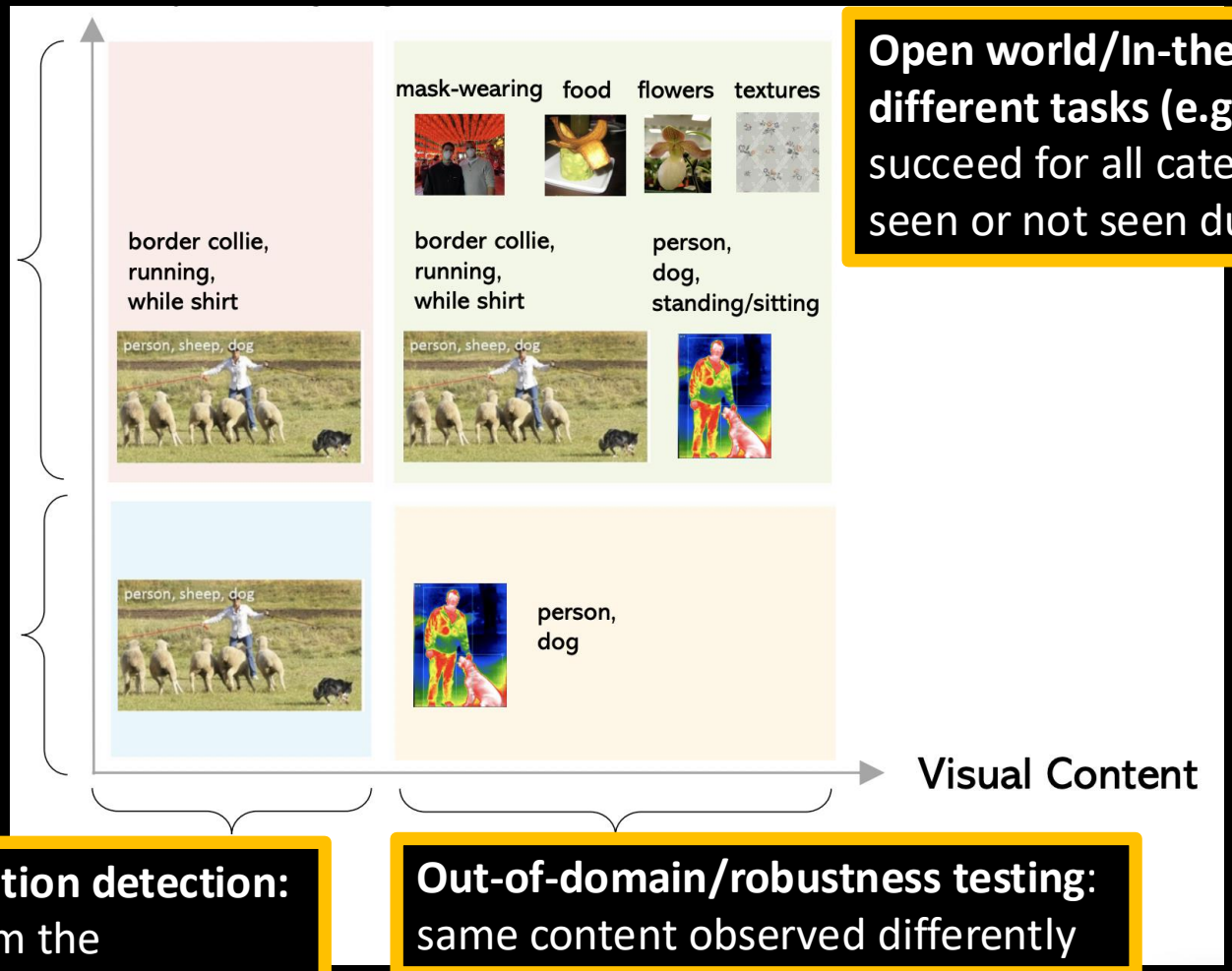
Bommasani et al. On the Opportunities and Risks of Foundation Models. arXiv 2021.

Surprise: Foundation Models Generalized As Is Beyond Closed-World Settings

Vocabulary / Language

Zero-shot and open vocabulary: generalize without labeled training data for target task (e.g., novel categories); the latter setting permits annotations with novel category (for a different task)

Closed-world/Closed-set: assumes all target classes are available at training



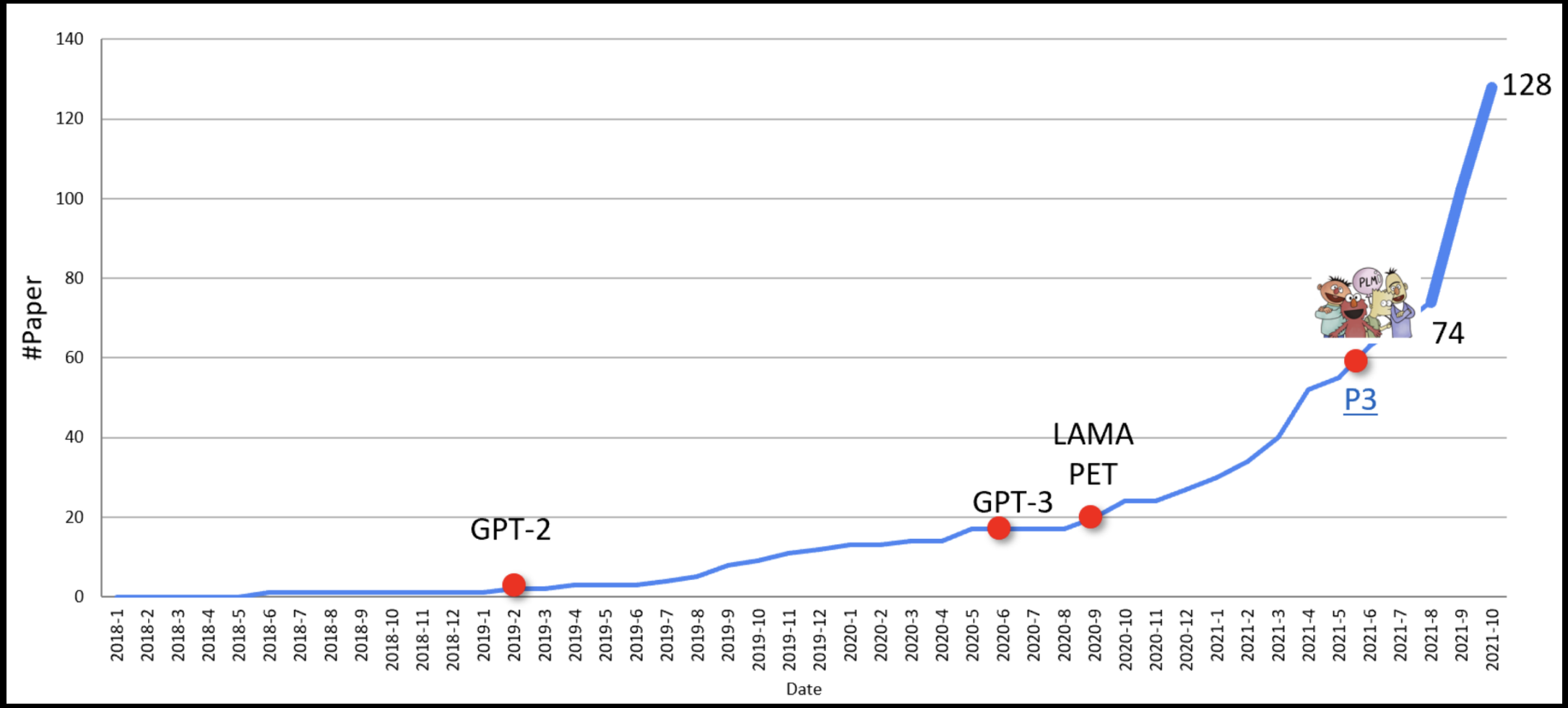
Open world/In-the-wild for different tasks (e.g., detection): succeed for all categories, whether seen or not seen during training

Open set classification/out-of-distribution detection: predict whether a sample is drawn from the distribution observed at training time

Out-of-domain/robustness testing: same content observed differently

https://arxiv.org/pdf/2210.09263.pdf

Growing Popularity of Prompt-based Research



Today's Topics

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- Thinking slower: reasoning for better outcomes
- Beyond text: prompting in other modalities
- Prompting activity

(ACM Computing Surveys, 2023)

Pre-train, Prompt, and Predict: A Systematic Survey of Prompting Methods in Natural Language Processing

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"as of this writing in 2021, we are in the middle of a second sea change, in which the "pre-train, fine-tune" procedure is replaced by one in which we dub "pre-train, prompt, and predict". In this paradigm… tasks are reformulated to look more like those solved during the original LM training with the help of a textual prompt... there is a catch – this method introduces the necessity for prompt engineering, finding the most appropriate prompt to allow a LM to solve the task at hand."

“Prompt engineering is the process of creating a prompting function fprompt(x) that results in the most effective performance on the downstream task.”- pg. 11

Liu et al. arXiv 2023

Manual Template Engineering: "human engineer or algorithm searches for the best template for each task the model is expected to perform"

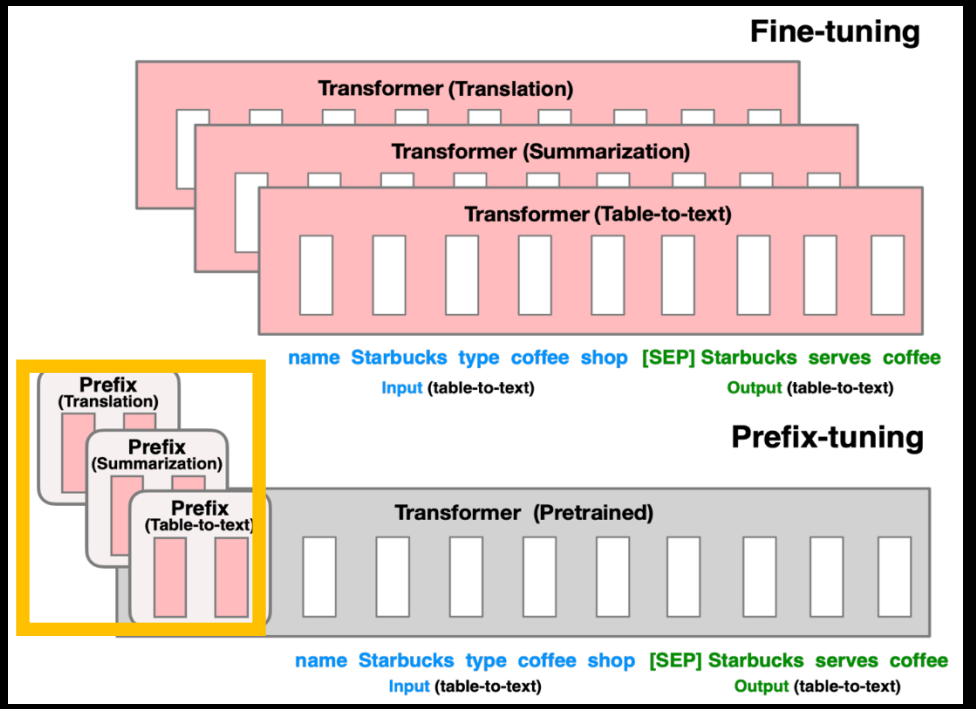
e.g.,

Task	Input ([X])	Template	Answer ([Z])
Sentiment	I love this movie.	[X] The movie is [Z].	great fantastic ...
Topics	He prompted the LM.	[X] The text is about [Z].	sports science ...
Intention	What is taxi fare to Denver?	[X] The question is about [Z].	quantity city ...
Aspect Sentiment	Poor service but good food.	[X] What about service? [Z].	Bad Terrible ...
NLI	[X1]: An old man with ... [X2]: A man walks ...	[X1]? [Z], [X2]	Yes No ...
NER	[X1]: Mike went to Paris. [X2]: Paris	[X1][X2] is a [Z] entity.	organization location ...
Summarization	Las Vegas police ...	[X] TL;DR: [Z]	The victim ... A woman ... ...
Translation	Je vous aime.	French: [X] English: [Z]	I love you. I fancy you. ...

Liu et al. arXiv 2023

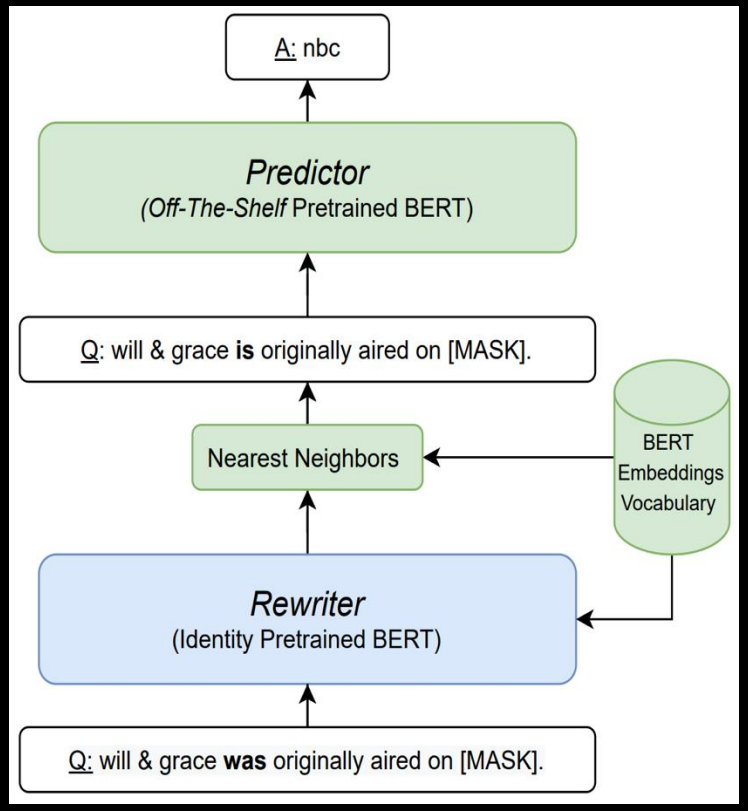
Automated Template Learning:

- continuous/soft prompts: tune a small number of additional parameters (rather than fine-tune the pretrained transformer); e.g.,



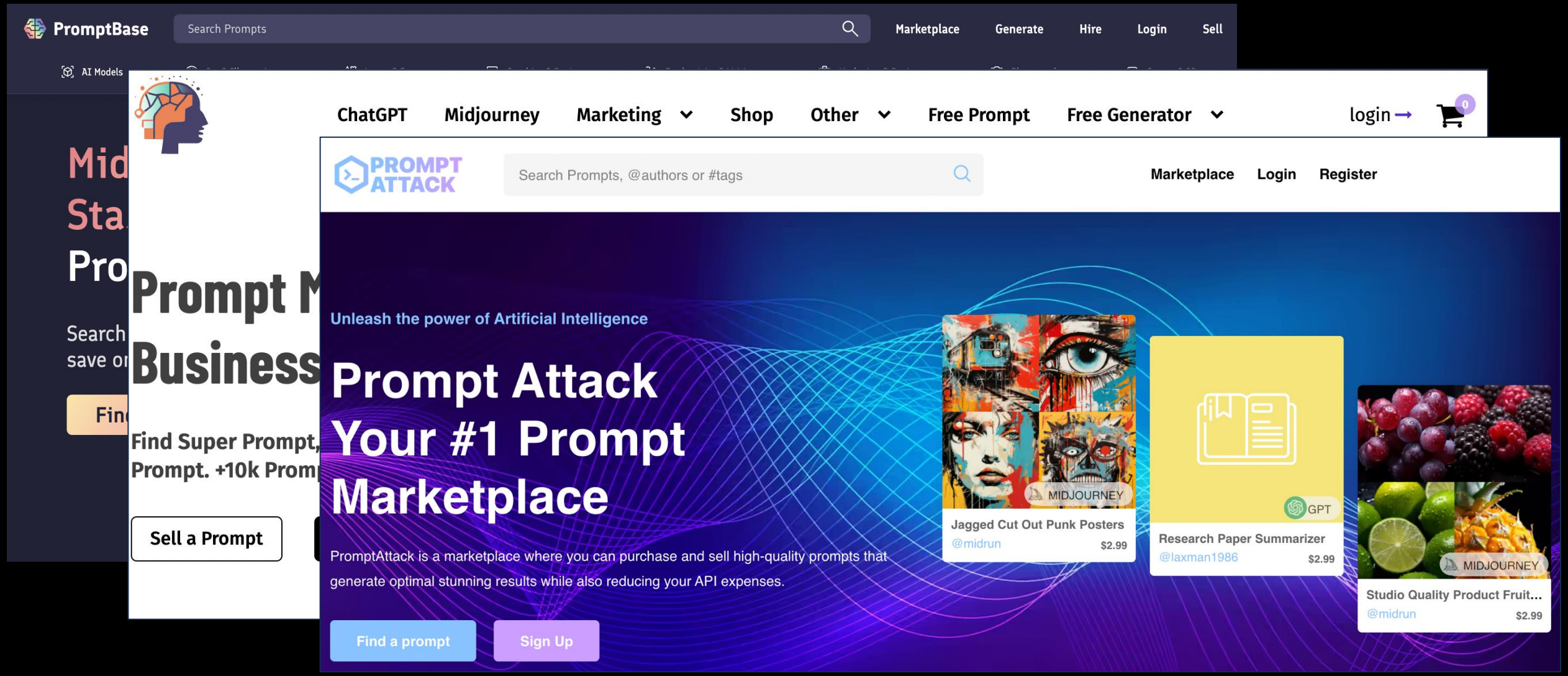
Li and Liang. ACL 2021

- discrete/hard prompts: e.g., paraphrases prompt to one that achieves higher accuracy



Haviv et al. ACL 2021

Many Companies Sold/Sell/Advertise Prompts



This Work Also Helped Inspire the Rise of Conversational Models; e.g.,

- Nov 2022: OpenAI's ChatGPT
- Feb 2023: Google's Gemini (originally Bard)
- Mar 2023: Anthropic's Claude

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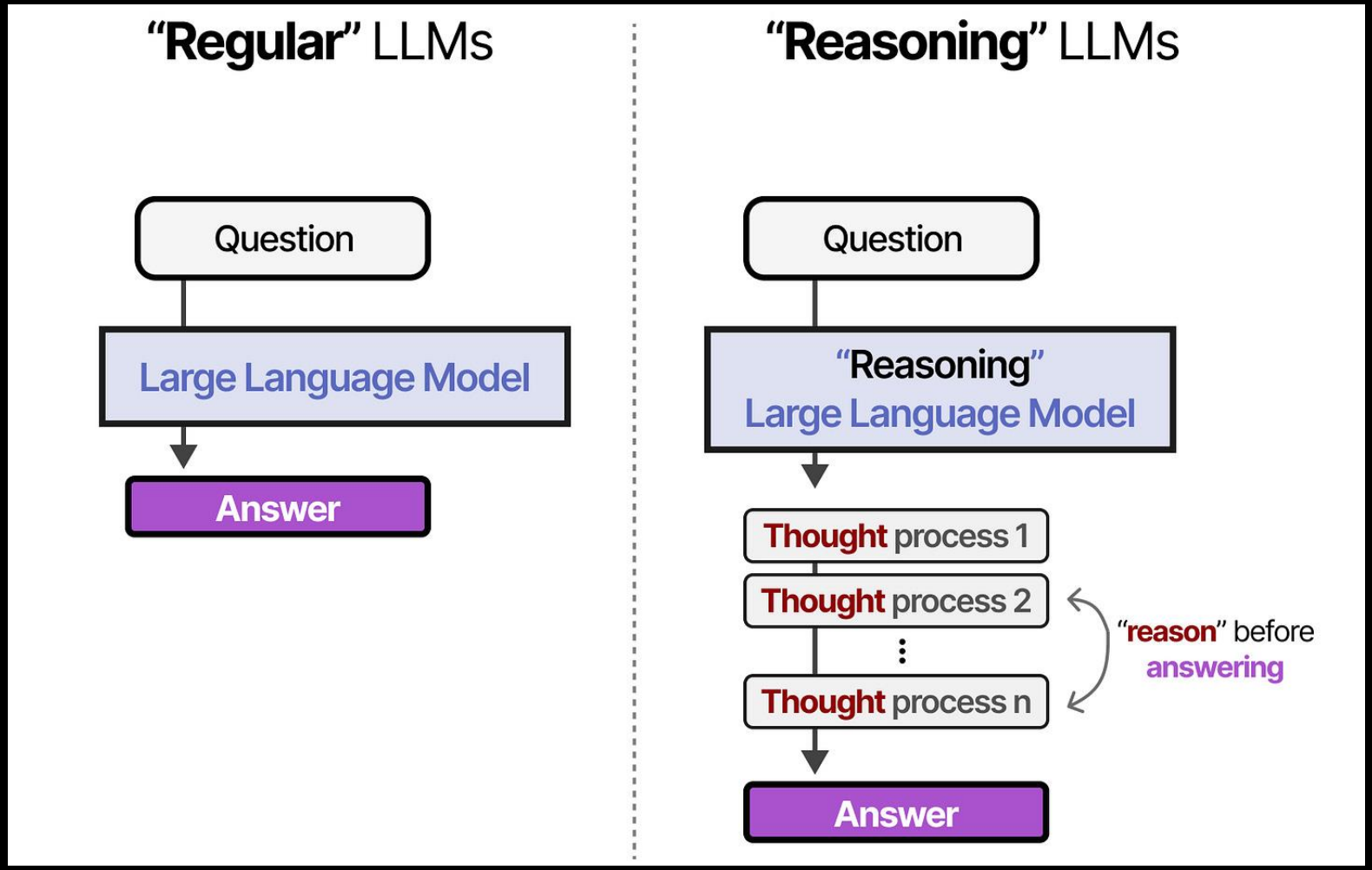
Intuition: How Much Reasoning Do You Do?

- What is 2 + 2?
  - 4

- What is 24 x 14?
  - My reasoning process:
  - 24 x 10 = 240
  - 24 x (14-10) = 24 x 4 = 96
  - 240+96 = 336 (final answer!)
  - Did you follow the same or a different reasoning path to generate the answer?

- Humans perform better when reasoning/thinking before responding

Key Finding: Models Perform Better When Encouraged to Reason/Think Before Responding



https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-reasoning-llms

(Pioneering paper, Neurips 2022)

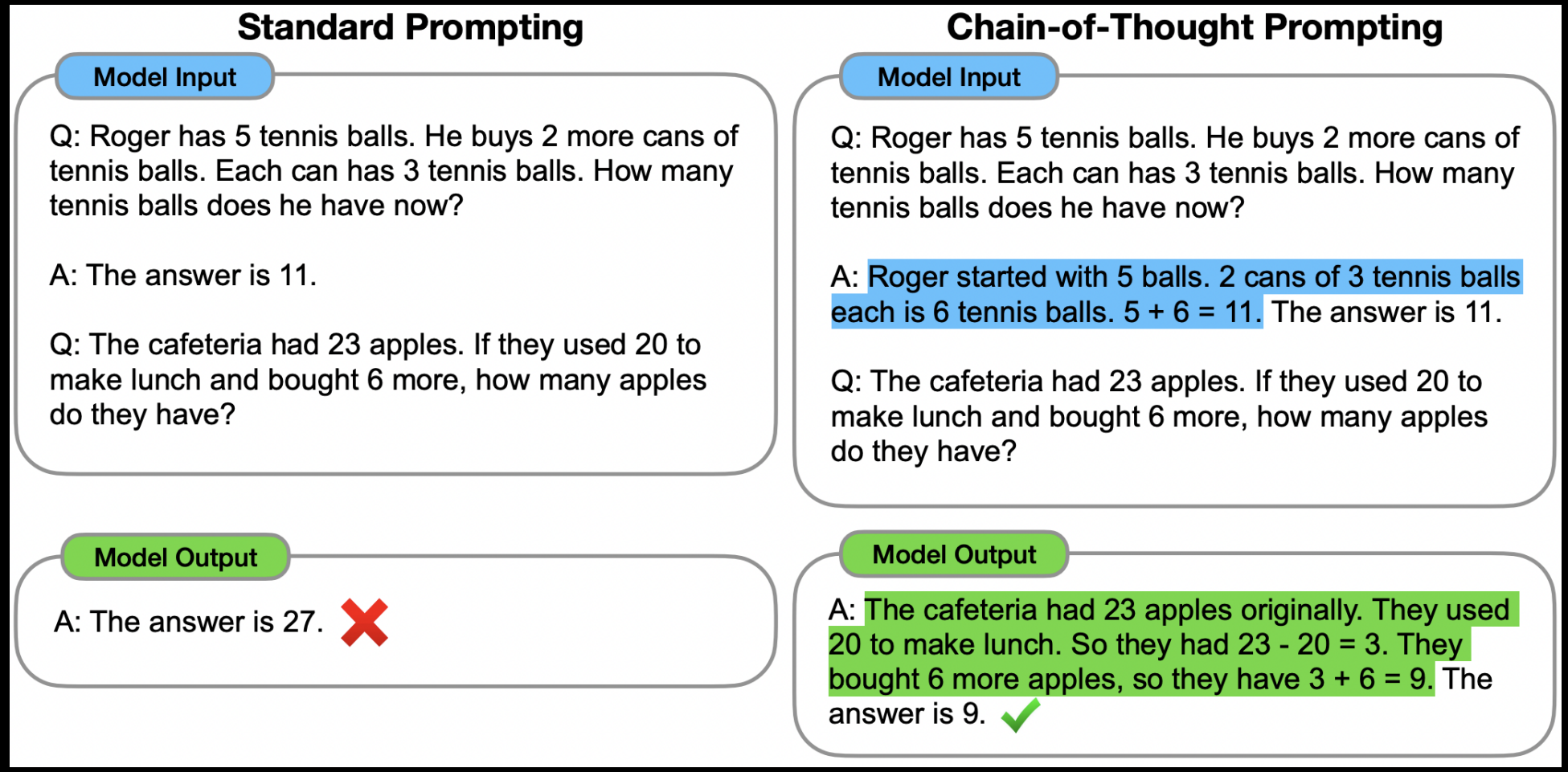
Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

Jason Wei **Xuezhi Wang** **Dale Schuurmans** **Maarten Bosma**

Brian Ichter **Fei Xia** **Ed H. Chi** **Quoc V. Le** **Denny Zhou**

Google Research, Brain Team

CoT Prompting



Guides model to show intermediate reasoning steps

Wei et al. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. Neurips 2022.

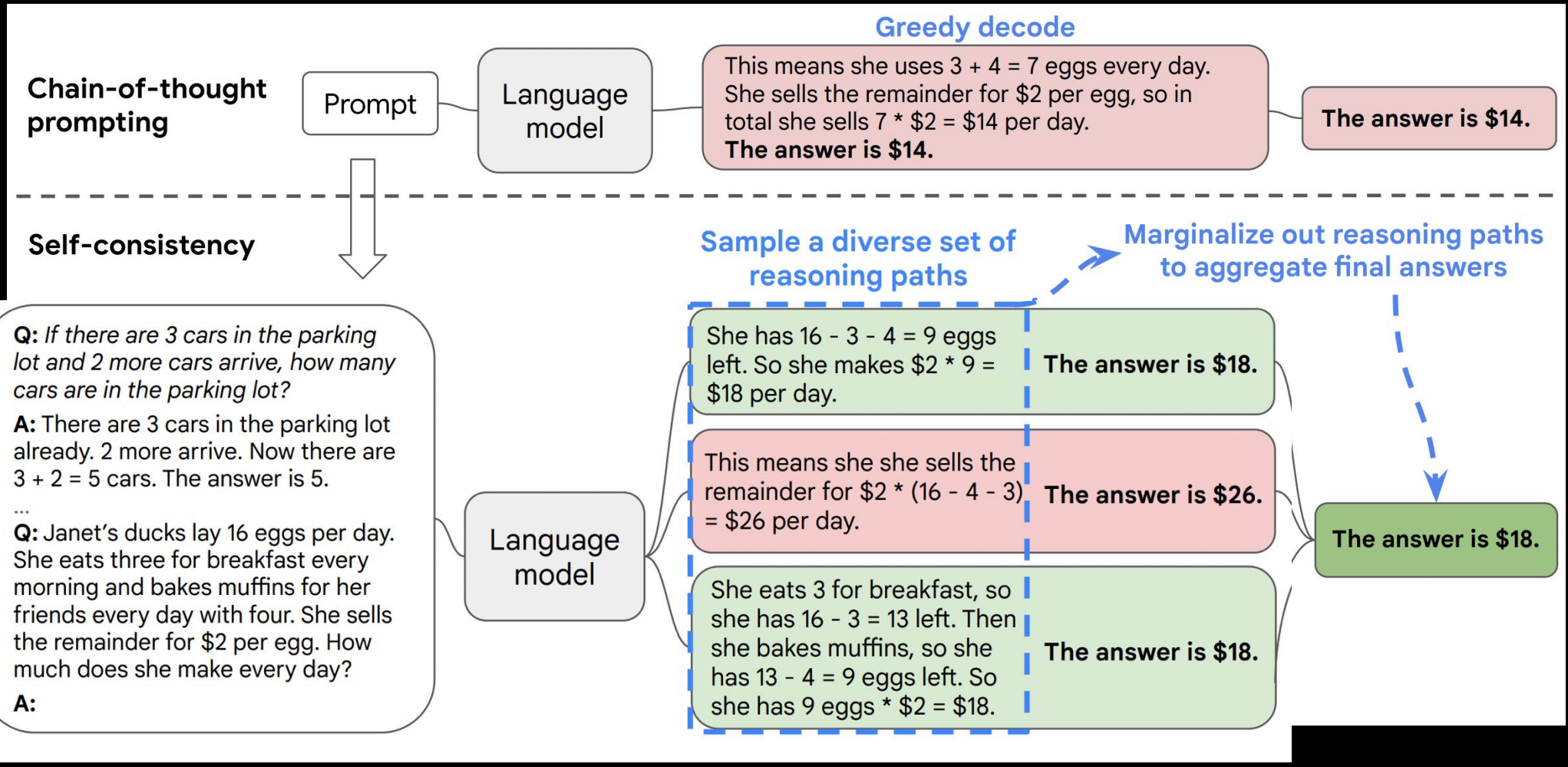
CoT Prompting: Why Does it Work?

CoT	Originally, Leah had 32 chocolates and her sister had 42. So in total they had 32 + 42 = 74. After eating 35, they had 74 - 35 = 39 pieces left in total. The answer is 39.	Julie is reading a 120-page book. Yesterday, she read 12 pages and today, she read 24 pages. So she read a total of 12 + 24 = 36 pages. Now she has 120 - 36 = 84 pages left. Since she wants to read half of the remaining pages, she should read 84 / 2 = 42 pages. The answer is 42. ✓
Invalid Reasoning	Originally, Leah had 32 chocolates and her sister had 42. So her sister had 42 - 32 = 10 chocolates more than Leah has. After eating 35, since 10 + 35 = 45, they had 45 - 6 = 39 pieces left in total. The answer is 39.	Yesterday, Julie read 12 pages. Today, she read 12 * 2 = 24 pages. So she read a total of 12 + 24 = 36 pages. Now she needs to read 120 - 36 = 84 more pages. She wants to read half of the remaining pages tomorrow, so she needs to read 84 / 2 = 42 pages tomorrow. The answer is 42. ✓

Benefit of examples appears to be revealing target format, as performance improves even with invalid examples

Wang et al. Towards Understanding Chain-of-Thought Prompting: An Empirical Study of What Matters. arXiv 2022.

CoT Prompting Enhancement: Self-Consistency

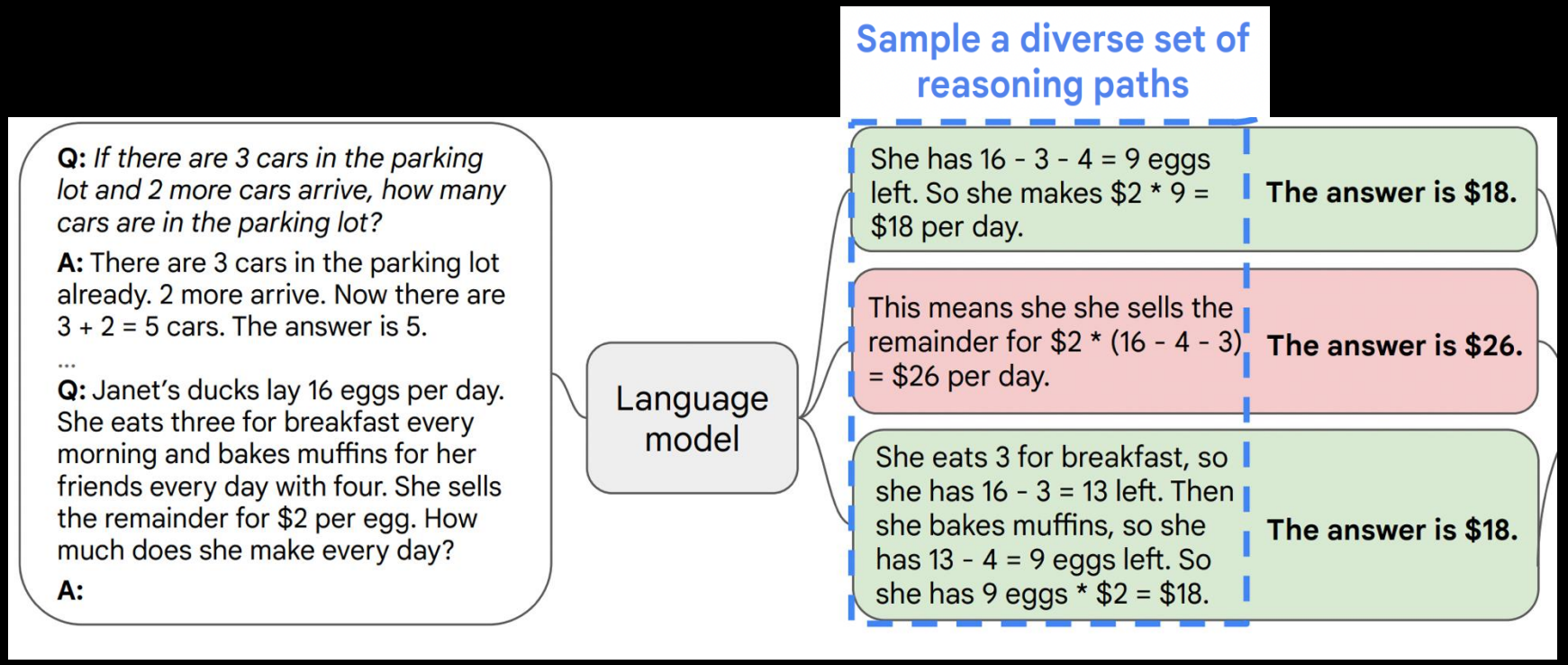


Use the most common response from multiple reasoning paths

Wang et al. Self-consistency Improves Chain of Thought Reasoning in Language Models. ICLR 2023.

CoT Prompting Enhancement: Self-Consistency

Introduces diversity by randomly sampling, at each time step, the next token from top *K* most probable tokens (from softmax layer)



Wang et al. Self-consistency Improves Chain of Thought Reasoning in Language Models. ICLR 2023.

CoT Variant

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

(Output) The answer is 8. ✗

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are 16 / 2 = 8 golf balls. Half of the golf balls are blue. So there are 8 / 2 = 4 blue golf balls. The answer is 4. ✓

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: The answer (arabic numerals) is

(Output) 8 ✗

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: ***Let's think step by step.***

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

Even when just prompted to reason, models perform better

(Examples from GPT-3)

Kojima et al. Large Language Models are Zero-Shot Reasoners. Neurips 2022.

This Work Helped Inspire the Rise of “Reasoning” Models; e.g.,

- Sep 2024: OpenAI’s 01, world’s first reasoning model
- Nov 2024: Alibaba’s QWQ, or Questions with Qwen *(model open-source; extends Meta’s Llama)*
- Dec 2024: Google’s Gemini Flash Thinking and OpenAI’s 03
- Jan 2025: DeepSeek’s R1 *(model open-source)*

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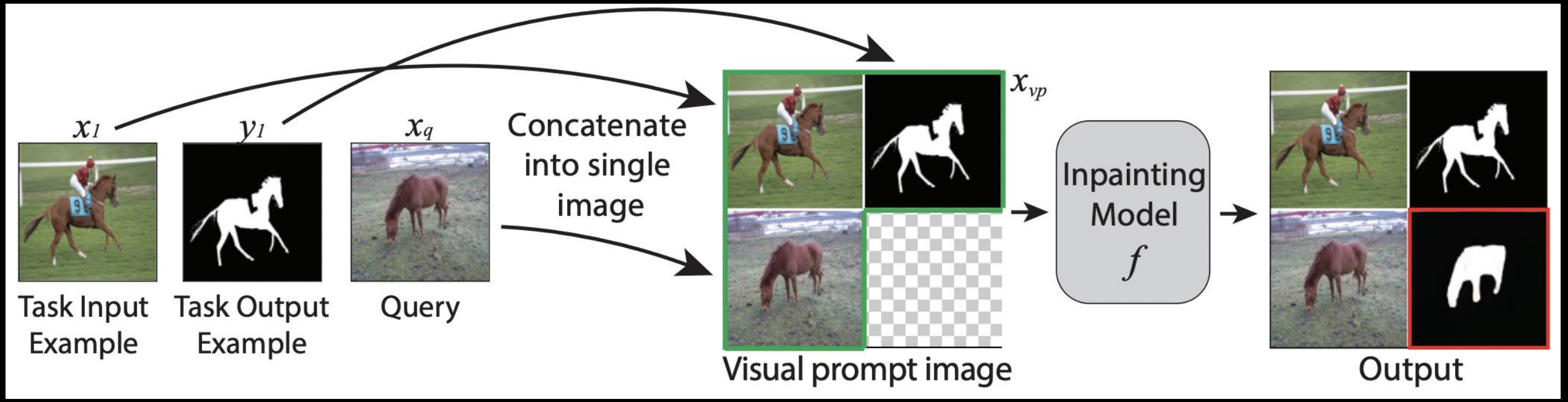
Motivating Observation

- Foundation models increasingly achieved state-of-the-art performance for NLP tasks with prompting:
  - Zero-shot
  - Few-shot using “in-context” examples; i.e., input: [task description, examples, prompt], such as “Translate English to Spanish. Computer -> Computadora. Vision”

- Idea: generalize these trends to other modalities

Large Vision Models (LVMs)

Visual Prompting as Image Inpainting

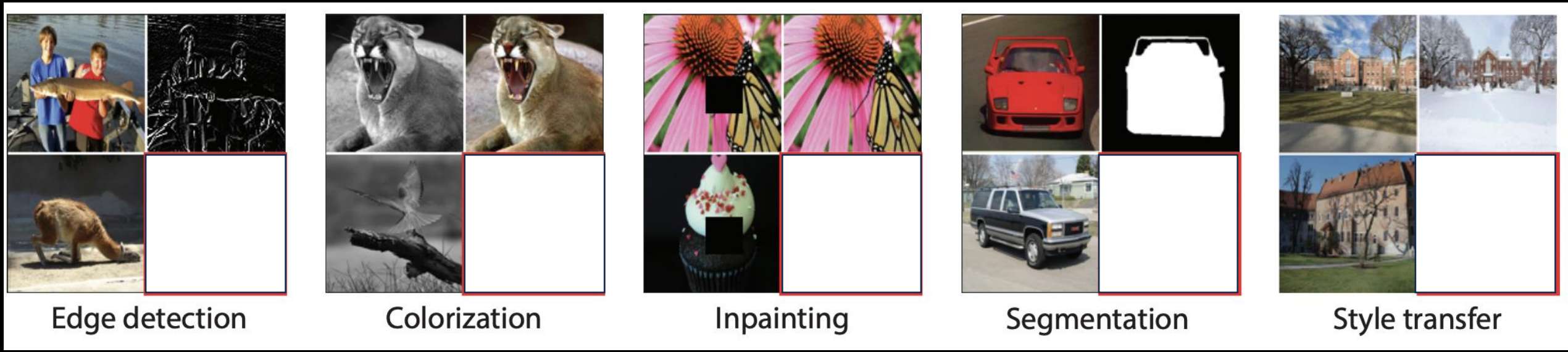


Designed to adapt to any “image-to-image translation” task by using the model as is (e.g., no fine-tuning required);

What should be predicted in this example?

Bar et al. Visual Prompting via Image Inpainting. Neurips 2022.

Visual Prompting as Image Inpainting

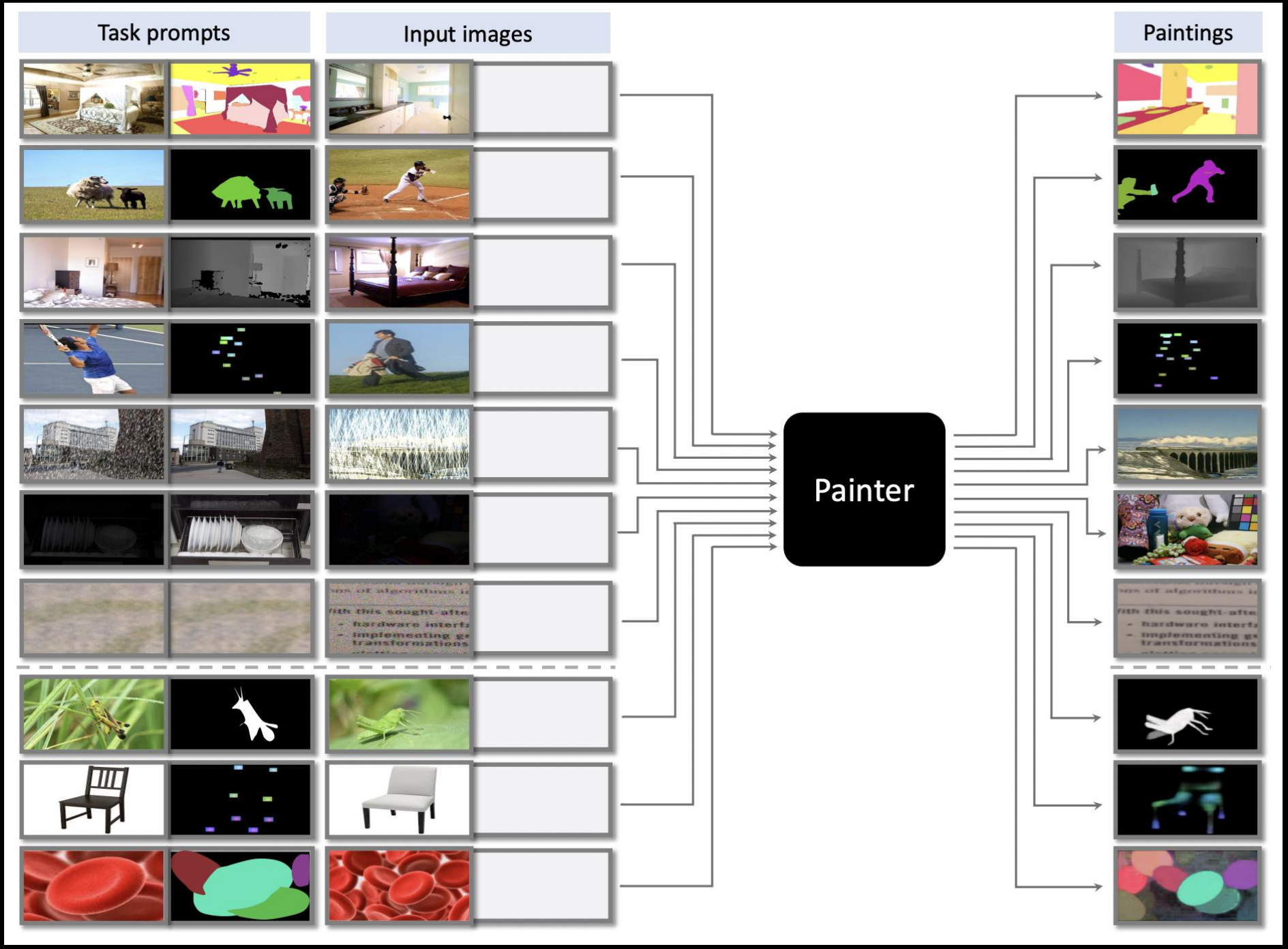


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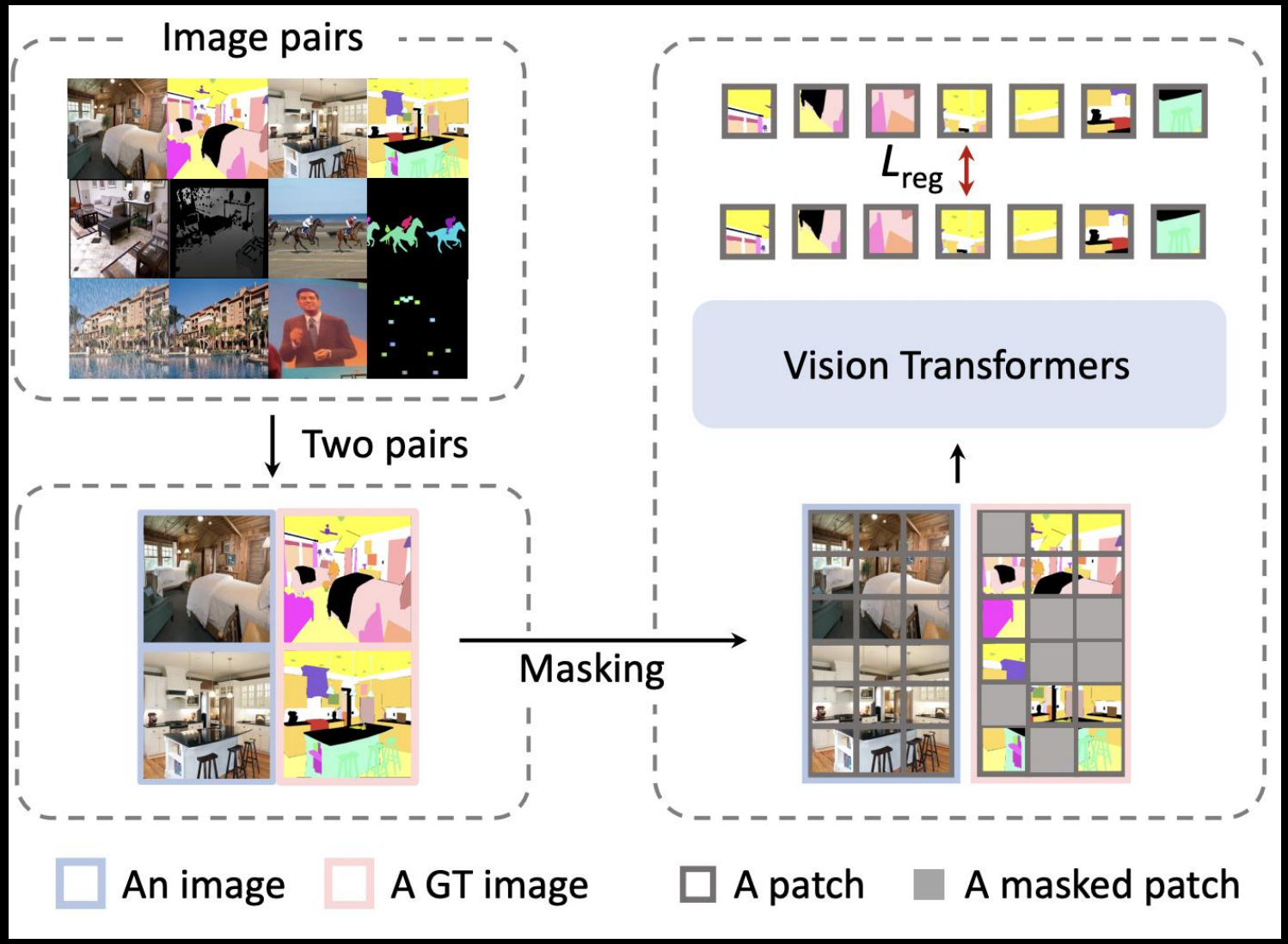
Visual Prompting

Extended in 2023 to standard vision benchmarks:



Wang et al. Images Speak in Images: A Generalist Painter for In-Context Visual Learning. CVPR 2023.

Training: Masked Image Modeling



Self-supervised learning where the model predicts values in masked out patches

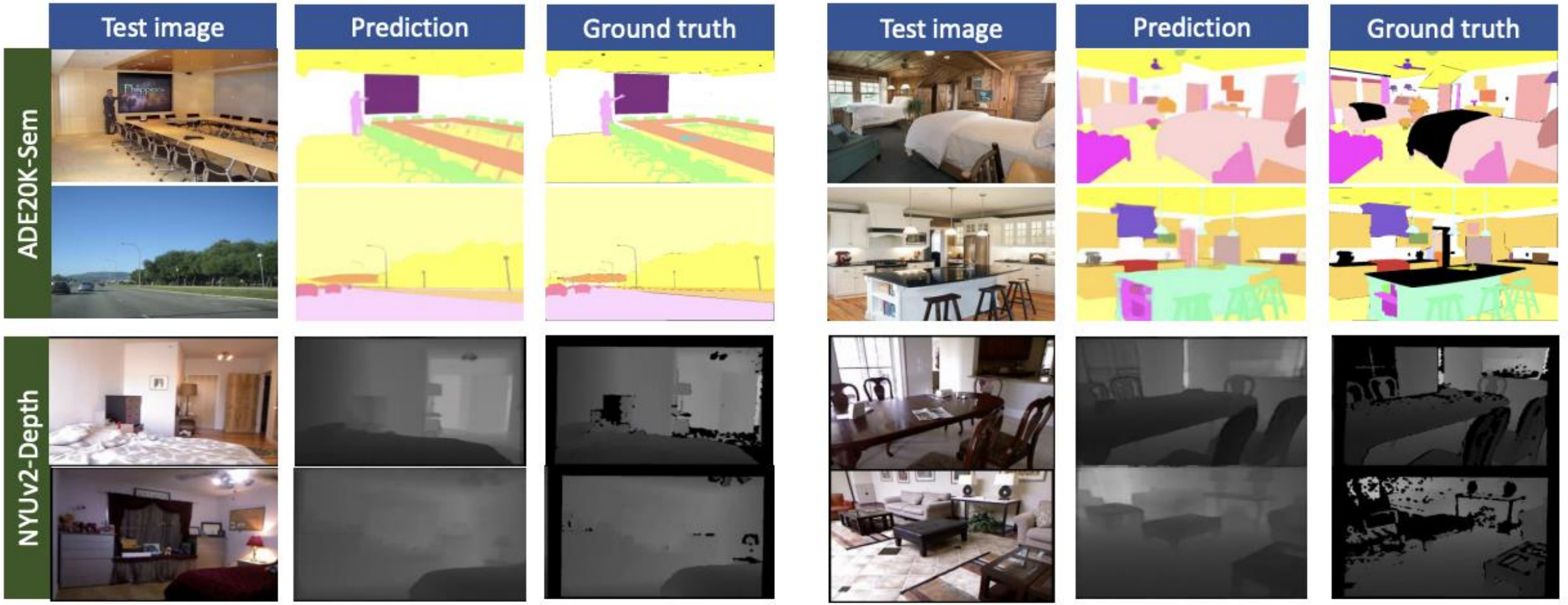
Wang et al. Images Speak in Images: A Generalist Painter for In-Context Visual Learning. CVPR 2023.

Experimental Results

Model achieved state-of-the-art performance on depth estimation for NYUv2 dataset and outperforms other generalist models on several more tasks.

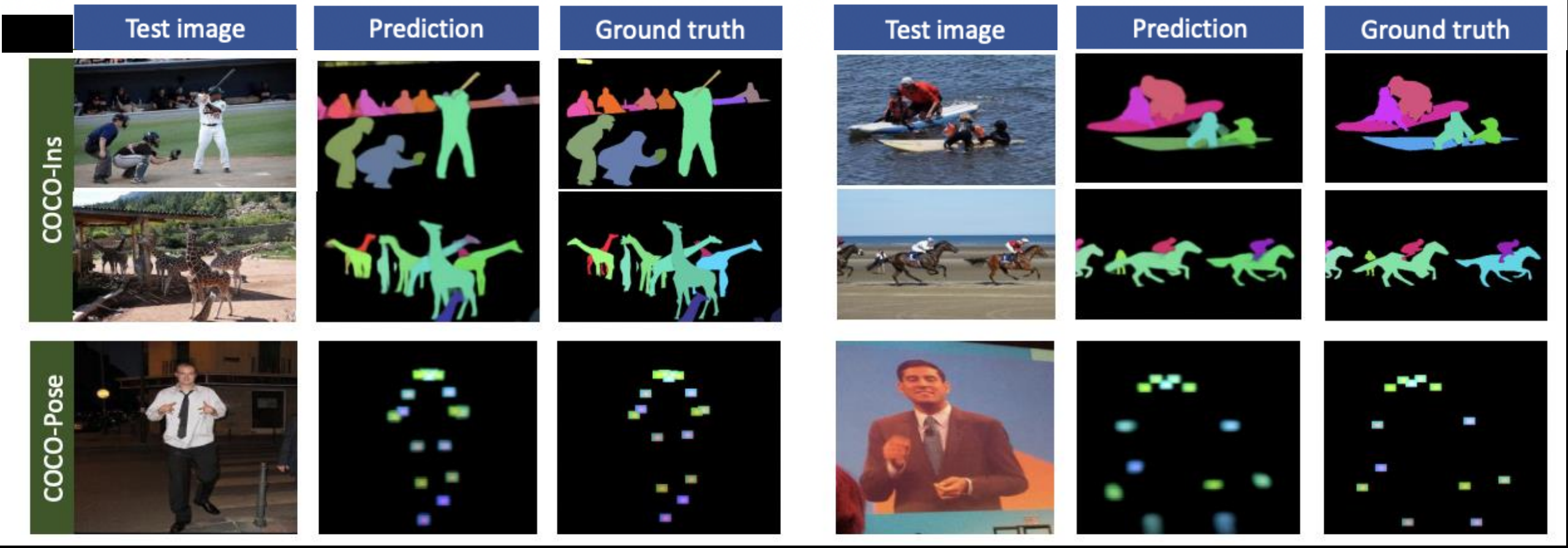
(Prompt used was best performing example pair per task from all examples in the training dataset)

Qualitative Results: In-Domain Results



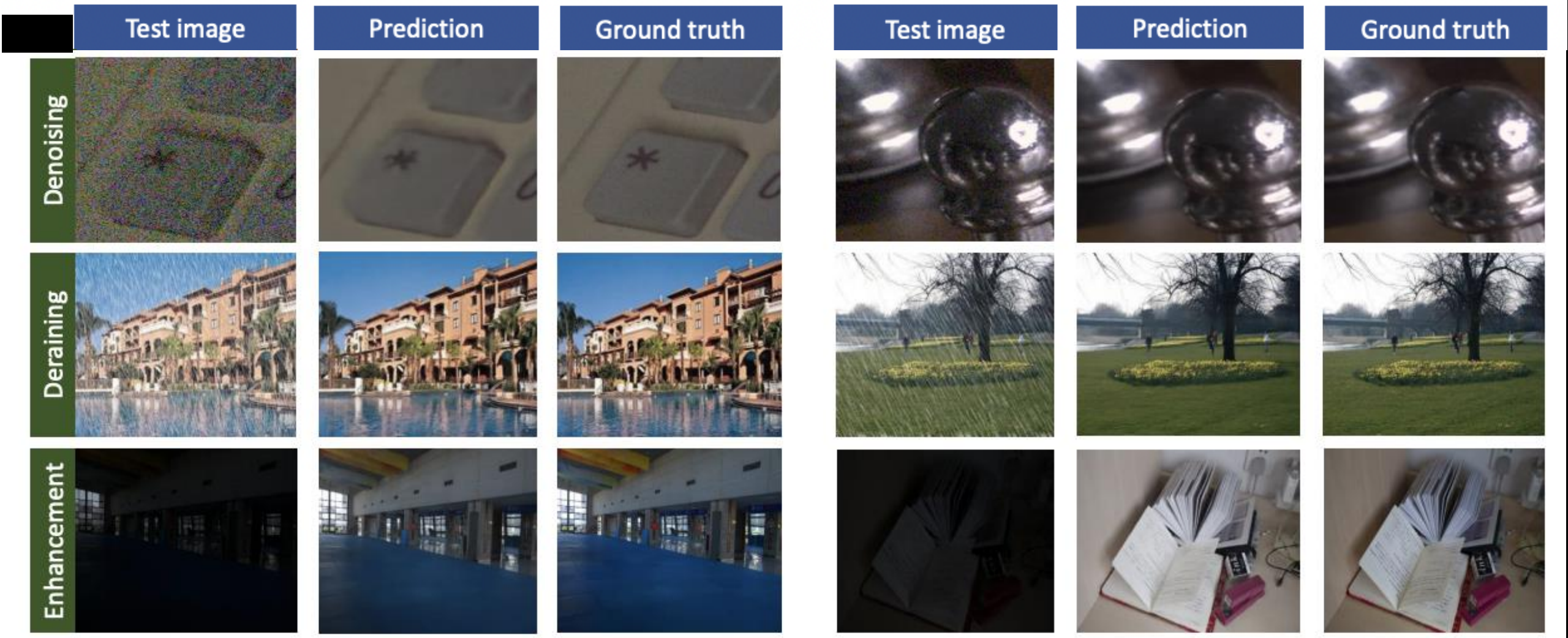
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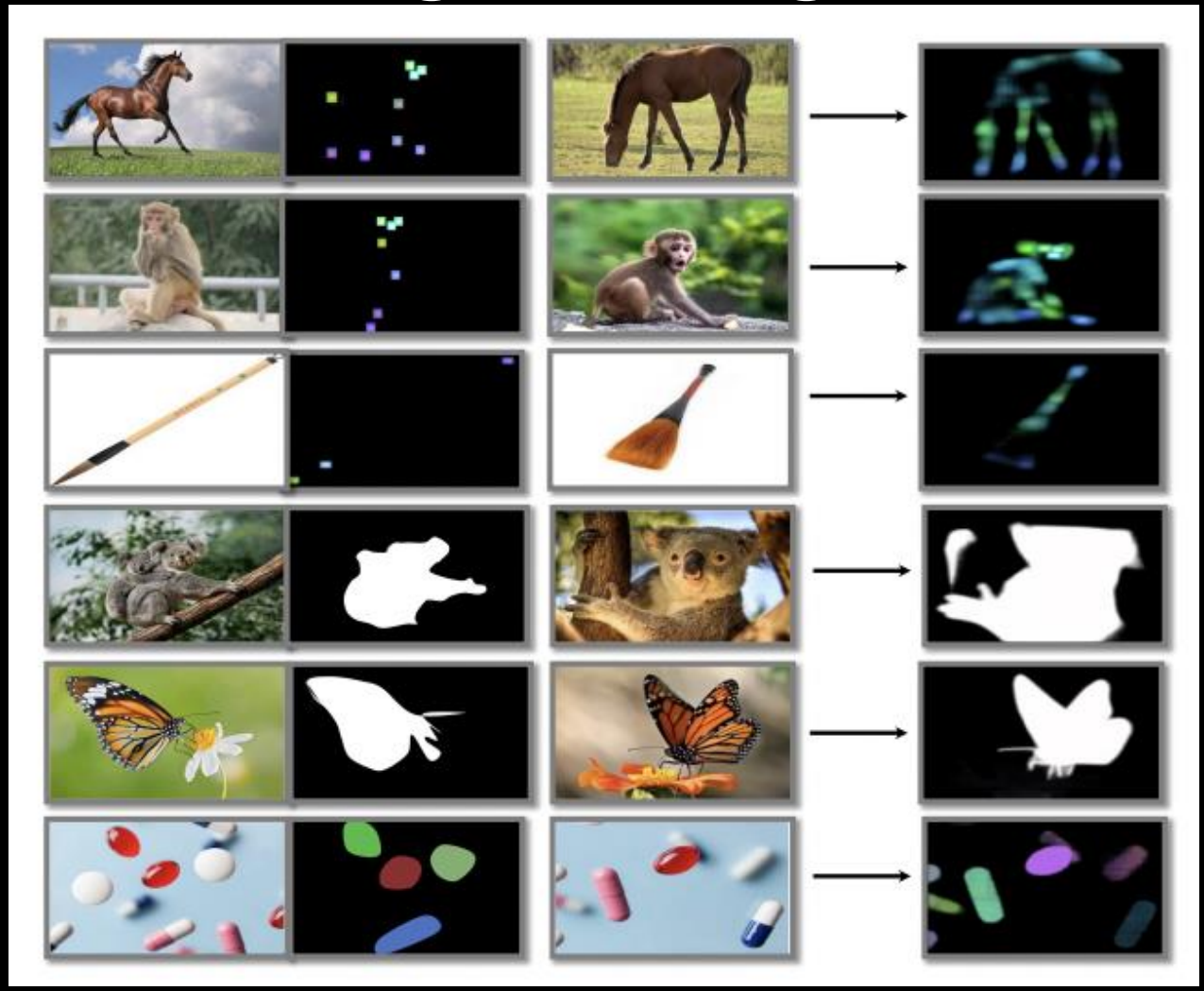
Wang et al. Images Speak in Images: A Generalist Painter for In-Context Visual Learning. CVPR 2023.

Qualitative Results: In-Domain Results



Qualitative Results: Open-Vocabulary Results (i.e., Categories Not Seen During Training)

Shows in-context examples, prompts, and predictions for keypoint detection, object segmentation, and instance segmentation



Wang et al. Images Speak in Images: A Generalist Painter for In-Context Visual Learning. CVPR 2023.

Prompt Types: Textual vs Visual

- When to prefer a visual prompt versus a textual prompt?
  - Visual offers greater equity for different languages, as non-English languages often are poorly supported if at all
  - Both can empower people appropriately based on their (dis)abilities: e.g., blind and deaf users

Large Vision-Language Models (VLMs) and Large Multimodal Models (LMMs) for Understanding and Generation

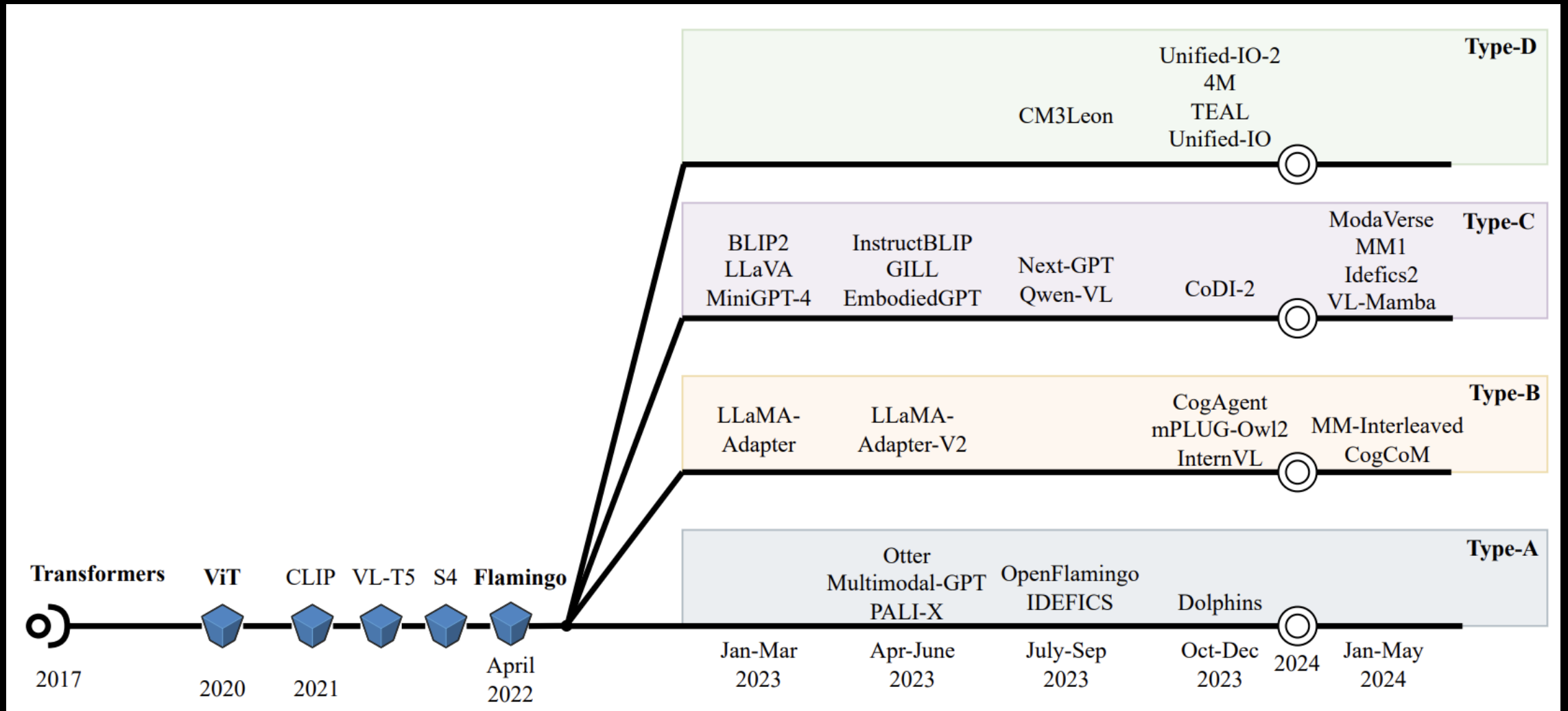


Figure 1: Development timeline of Multimodal models grouped in four proposed architecture types.

Wadekar et al. The Evolution of Multimodal Model Architectures. arXiv 2024.

(Pioneering LMM; Neurips 2022)

Flamingo: a Visual Language Model for Few-Shot Learning

Jean-Baptiste Alayrac[*,‡] Jeff Donahue[*] Pauline Luc[*] Antoine Miech[*]

Iain Barr[†] Yana Hasson[†] Karel Lenc[†] Arthur Mensch[†] Katie Millican[†]

Malcolm Reynolds[†] Roman Ring[†] Eliza Rutherford[†] Serkan Cabi Tengda Han

Zhitao Gong Sina Samangooei Marianne Monteiro Jacob Menick

Sebastian Borgeaud Andrew Brock Aida Nematzadeh Sahand Sharifzadeh

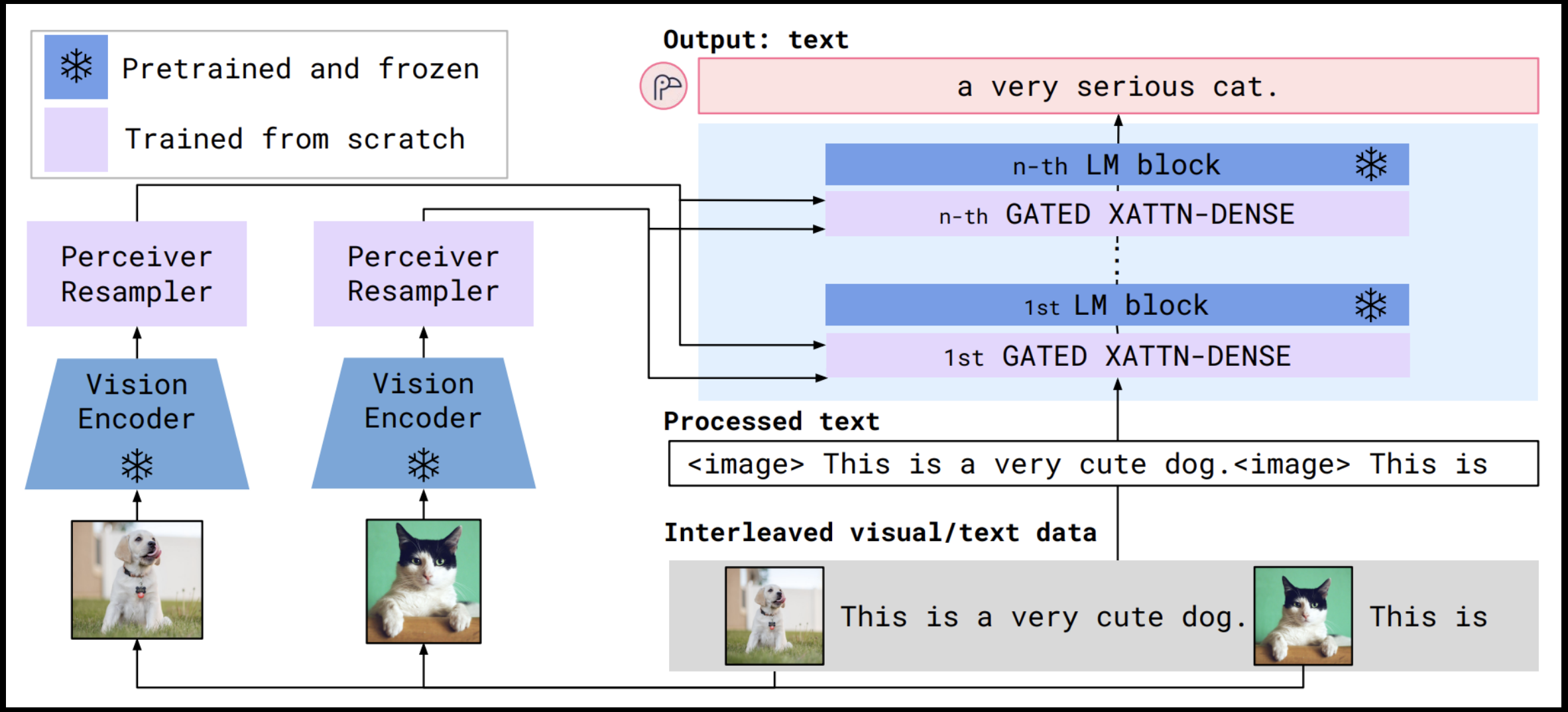
Mikolaj Binkowski Ricardo Barreira Oriol Vinyals Andrew Zisserman

Karen Simonyan[*,‡]

[*] Equal contributions, ordered alphabetically, [†] Equal contributions, ordered alphabetically,
[‡] Equal senior contributions

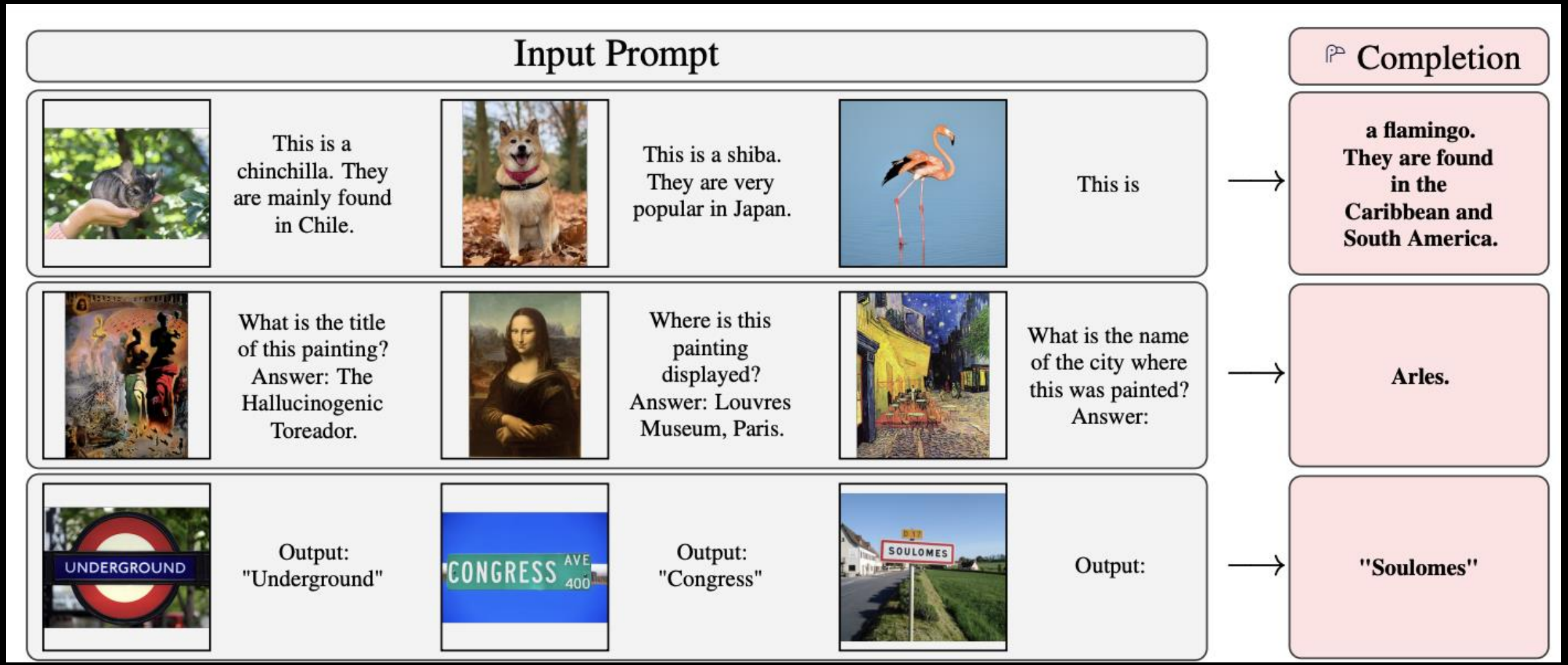
DeepMind

Key Challenge: Interleaved Image-Text Input



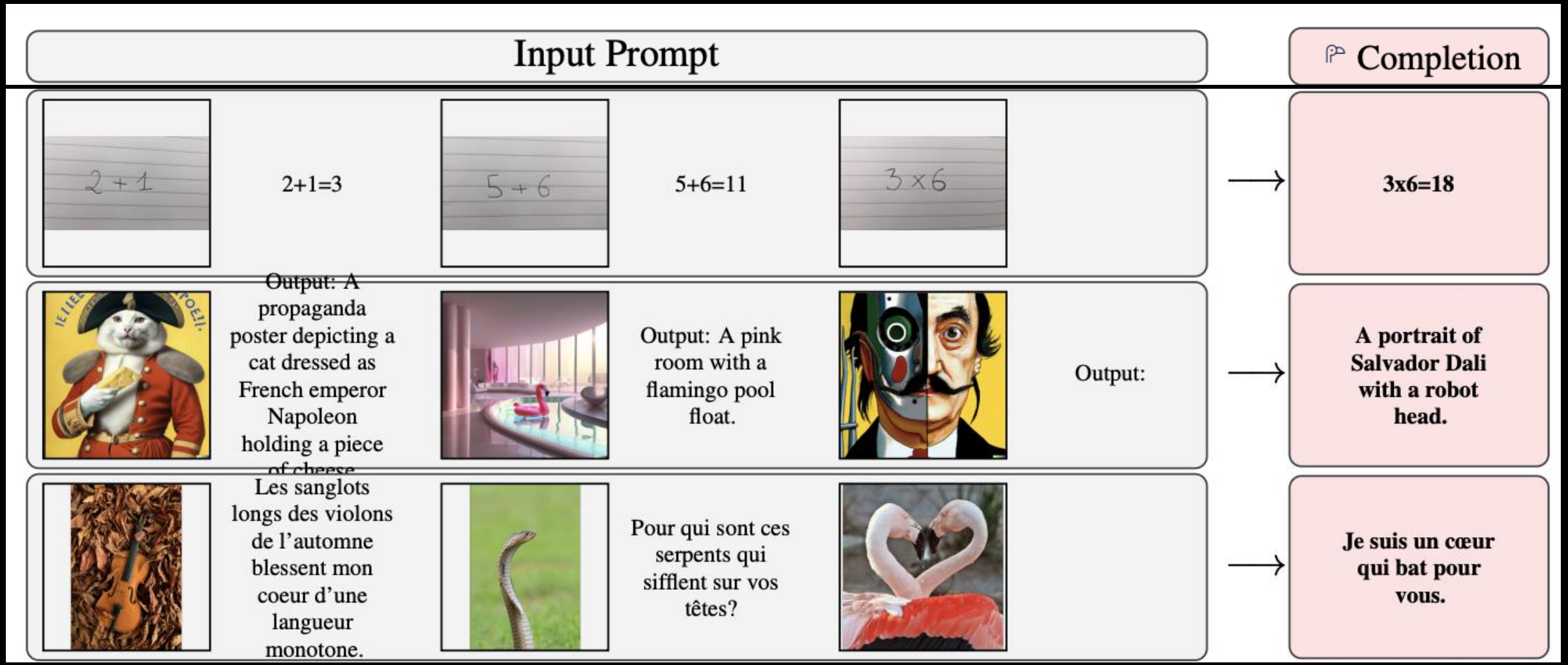
Alayrac et al. Flamingo: A Visual Language Model for Few-Shot Learning. Neurips 2022.

Achieved Strong Zero-Shot Performance with Multimodal Prompts for Many Tasks; e.g.,



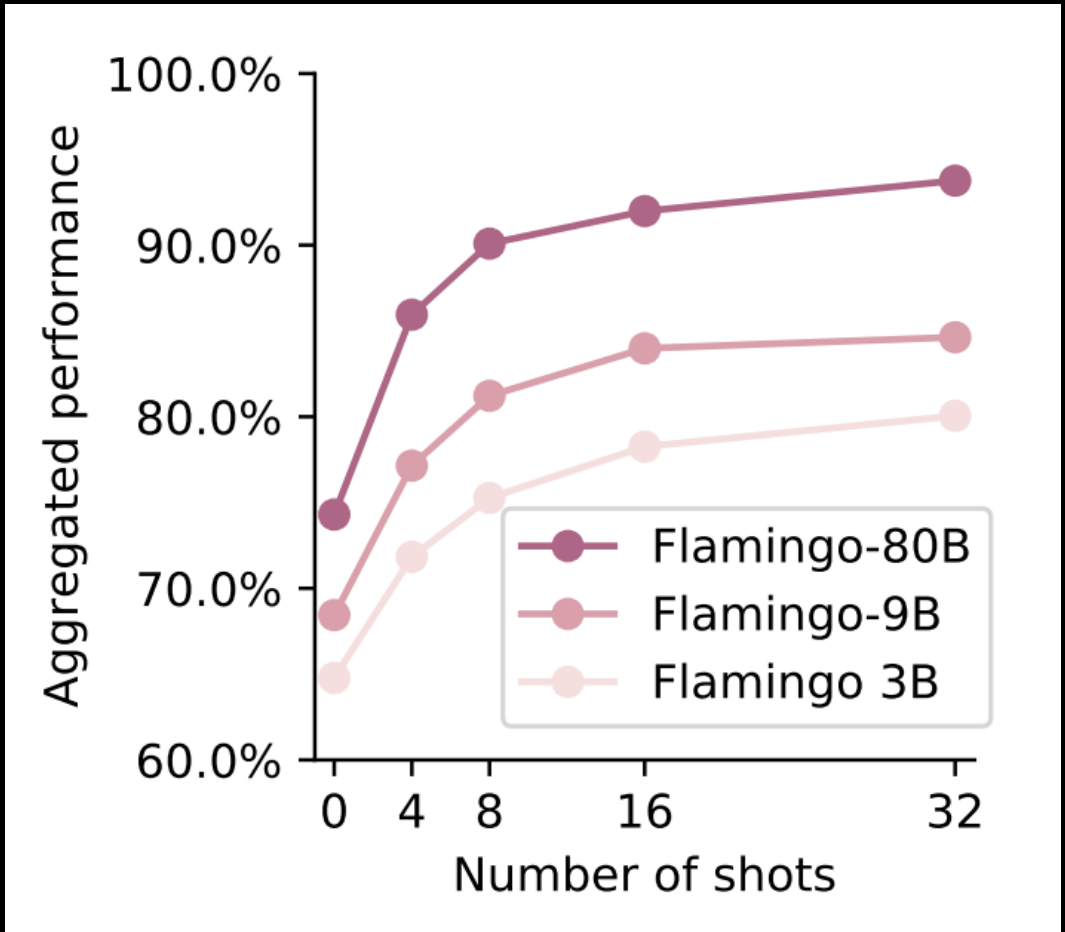
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Experimental Results

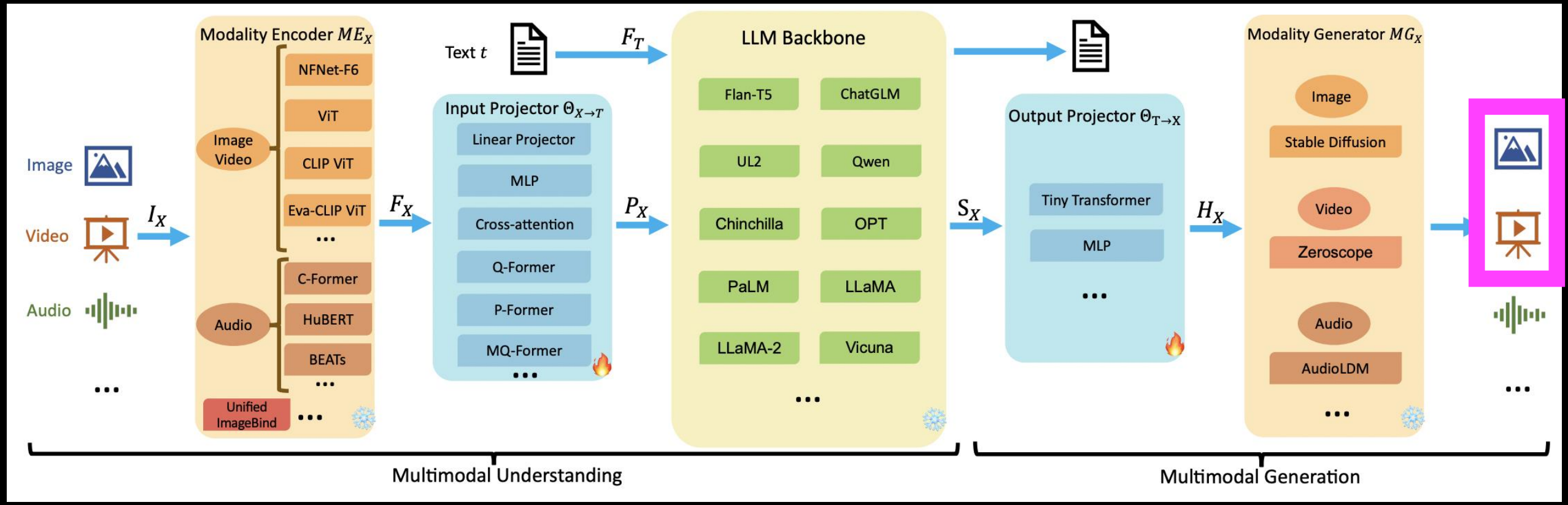


Largest model achieved state-of-the-art on 6 of 16 tasks, outperforming fine-tuned models

What are trends for different model sizes and numbers of “shots”?

Alayrac et al. Flamingo: A Visual Language Model for Few-Shot Learning. Neurips 2022.

Multimodal Understanding vs Generation



e.g., many methods convert non-textual inputs to text descriptions that LLMs understand (i.e., prompts) or finetune the vision encoder to align with a frozen pre-trained LLM

e.g., largely based on diffusion models, which are trained to recover the original content from noise

Zhang et al. MM-LLMs: Recent Advances in MultiModal Large Language Models. arXiv 2024.

Pioneering LMMs

- Industry examples for vision + language
  - OpenAI's GPT-4(Vision)
  - Google's Gemini

- Beyond VLMs; e.g.,
  - PandaGPT: text, image, video, audio, and sensors for depth (3D), thermal (infrared radiation), or inertial measurement units (IMU) of motion and position; connects objects with sound, 3D shape, temperature, and motion
  - SpeechGPT: language and speech
  - NExT-GPT: text, images, videos, and audio

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